

```

i=4*r+c;
getImage(s[i].left+15, s[i].
top+15, s[i].right, s[i].bottom,
vbuff1);
temp=s[i].valflag;
k=1;
--c;
i=4*r+c;
getImage(s[i].left+15, s[i].
top+15, s[i].right, s[i].bottom,
vbuff2);
putimage(s[i].left+15, s[i].
top+15, vbuff1,0);
s[k].valflag = s[i].valflag;
s[i].valflag= temp; /* copy lo-
cation of iflag image */
r++;
i=4*r+c;
putimage(s[i].left+15, s[i].
top+15, vbuff2,0);
break;

case 77 : /*right arrow */
dis_rem_moves();
if(c == 3)
{
setviewport(200,350,550,
360,0);
outtext("Not allowed use other
than right arrow");
vbell();
delay(200);
clearviewport();
setviewport(0,0,getmaxx(),getm
axy(),0);
c=3;
break;
}
/* other processing */
++c;
i=4*r+c;
temp=s[i].valflag;
k=1;
/* copy location of iflag image */

getImage(s[i].left+15,
s[i].top+15, s[i].right, s[i].
bottom,vbuff1);
--c;
i=4*r+c;
s[k].valflag = s[i].valflag;
s[i].valflag= temp; /* copy
location of iflag image */

getImage(s[i].left+15,
s[i].top+15, s[i].right, s[i].
bottom,vbuff2);
putimage(s[i].left+15, s[i].
top+15, vbuff1,0);
++c;
i=4*r+c;
putimage(s[i].left+15, s[i].
top+15, vbuff2,0);
break;

case 75 : /*left arrow */
dis_rem_moves();
if(c <= 0)
{
setviewport(200,350,550,
360,0);
outtext("Not allowed use other
than left arrow");
vbell();
delay(200);
clearviewport();
setviewport(0,0,getmaxx(),getm
axy(),0);
c=0;
break;
}
}
/* other processing */
--c;
i=4*r+c;
temp=s[i].valflag;
k=1;
/* copy location of iflag image */
putimage(s[i].left+15, s[i].
top+15, vbuff1,0);
s[k].valflag = s[i].valflag;
s[i].valflag= temp; /* copy lo-
cation of iflag image */
c--;
i=4*r+c;
putimage(s[i].left+15, s[i].
top+15, vbuff2,0);
break;

case 1: /* ESC key */
break;
default: vbell(); break;
}
/* to check each movement with
a standard pattern */
result=0;
match_result(s);
if(result == 1 && jmoves <
50)
{
settextstyle(0,0,02);
outtextxy(180,400,"YOU WON THE
GAME !");
settextstyle(0,0,01);
break;
}
else
if(result== 0 && jmoves >=
50)
{
settextstyle(0,0,02);
outtextxy(180,400,"THE GAME
OVER ! YOU LOST");
settextstyle(0,0,01);
break;
}
}

void vbell(void)
{
sound(2000);
delay(100);
nosound();
}

void flag_india(square sq[])
{
int gap = 5;

for(int i = 0; i<16; i++)
{
setcolor(62);
rectangle(sq[i].left+gap*2, sq[i].
top+gap*2, sq[i].right+gap, sq[i].
bottom+gap);
setcolor(61);
bar(sq[i].left+15, sq[i].top+15,
sq[i].right, sq[i].bottom);
setcolor(60);
draw_flag(s);
}
void draw_flag(square sq[])
{
int gap = 5;
/* fill flag color */
setfillstyle(01,60);
bar((sq[0].left+25), (sq[0].
top+21), (sq[3].right-10), (sq[3].bot-
tom-10));
/* white bar */
setfillstyle(01,63);
bar(sq[4].left+25, sq[4].top+10,sq[7].
right-10, sq[7].bottom+5);
/* green bar */
setfillstyle(01,02);
bar(sq[8].left+25, sq[8].top,sq[11].
right-10, sq[11].bottom-10);
/* pole color bar */
setfillstyle(01,20);
bar(sq[0].left+20, sq[0].top+20,sq[12].
left+25, sq[12].bottom-10);
/* ashoka chakra */
for(int i=0;i<360;i=i+20)
{
arc(midx+6,midy-
10,i,i+20,sq[5]/2-5);
setcolor(56);
{
int j=i+20;
sqx= midx+6 + (sq[5]/2-
5)*cos(j);
sqy = midy-10 + (sq[5]/2-
5)*sin(j);
line(midx+6,midy-10,sqx,sqy);
}
circle(midx+6,midy-10,sq[5]/2-5);
}
/* save flag interns of blocks */
for(i=0; i<16; i++)
{
sq[i].valflag=i;
getImage(sq[i].left+gap*2, sq[i].
top+gap*2, sq[i].right+gap, sq[i].
bottom+gap,
sq[i].imagflag);
}
void shuffle_flag(square sq[])
{
int j, gap = 5;
for(int i=0; i<16; i++)
{j= s[i].valflag;
putimage(sq[i].left+gap*2, sq[i].
top+gap*2, sq[j].imagflag,0);
}
outtextxy(260,340,"YOUR FLAG SCREEN");
settextstyle(0,0,0);
setcolor(04);
outtextxy(200,360," <Press any key to
return to main >");
getch();
clearviewport();
}
void match_result( square sq[])
{
for(int i = 0; i<16; i++)
{
if(sq[i].valflag != matchnb(i))
{result=0; break;}
else
{result=1;
continue;}
}
}
}

```